

Aligned but Inert: Concept Bottlenecks in Dense Prediction, and How to Make Them Causal

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Abstract. Concept bottleneck models promise intrinsic interpretability: route the prediction through a human-readable concept layer, and the concept activation *is* the explanation. We show that transplanting this idea into dense prediction fails silently, diagnose why, and repair it. We build Clinical Concept Routing (CCR), a bottleneck that assigns every encoder token to one of K clinically-labeled experts, and train it for 3D brain-tumor segmentation. By every measure this literature reports, it succeeds: routing tracks the clinical concepts at CAS_{fg} up to 0.96 and separates necrosis, edema and enhancing tumor at AUROC 0.88, against 0.74 for the strongest post-hoc attribution. Yet the routing is *causally inert*. Forcing tokens through a different expert changes 0.2% of voxel labels, and zeroing the entire bottleneck costs 1.5 Dice. Alignment metrics cannot detect this, because alignment is compatible with a bottleneck the model never consults—and the alignment loss trains precisely what those metrics score. We trace the failure to two leaks that encoder–decoder architectures invite, a residual path *through* the module and skip connections *around* it, and rule out the obvious explanation: the experts are not degenerate (0.9 pairwise divergence). A linear probe on an encoder trained with no concept supervision already reaches 0.81, so concept-discriminability gains over saliency baselines are weaker evidence than they appear. Attempts to force dependence by *removing* the decoder’s alternatives fail—the optimizer compensates. Making the routing an *additive term of the output* instead succeeds: causal control rises from 0.011 to **0.636** [0.621, 0.651] label-flip under intervention ($p < 10^{-28}$), removing the routing term costs up to 99% Dice, and the price is ≈ 1 Dice point. **A concept layer is an explanation only if the model consults it, and showing that requires intervening, not correlating.**

Keywords: Interpretability · Concept bottlenecks · Causal intervention · Medical image segmentation · Mixture-of-experts

1 Introduction

Deep segmentation models match expert radiologists on brain-tumor benchmarks, but clinical adoption lags because their decisions cannot be verifiably explained. The dominant remedy, post-hoc attribution [10,12,6], produces a

saliency map *after* the prediction and therefore approximates the mechanism rather than reporting it. Rudin [9] argues the alternative: build models whose decisions are interpretable by construction. Concept Bottleneck Models [5] realize this for classification by forcing the prediction through a human-readable concept layer, so the concept activation *is* the explanation.

Extending that idea to dense prediction is attractive and, we will show, treacherous. We build the natural extension—**Clinical Concept Routing (CCR)**: replace the encoder bottleneck of a segmentation network with a router that assigns every token to one of K clinically labeled experts {Background, Necrotic core, Edema, Enhancing tumor}. The routing distribution is a per-region, concept-labeled map, computed in one forward pass, with no post-hoc step. It looks exactly like an intrinsic explanation should.

It aligns beautifully, and it does nothing. Trained end-to-end, CCR’s routing tracks the clinical concepts closely (CAS_{fg} up to 0.96) and discriminates them far better than any post-hoc method (0.88 vs. 0.74). Both are the metrics this literature reports, and by both the module is a success. But when we *intervene*—force the tokens the router assigned to necrosis through the edema expert, changing nothing else—the prediction does not move: the target posterior shifts by 0.001 and 0.2% of voxels change label. Zeroing the bottleneck entirely costs 1.5 Dice. The routing is a well-trained annotation that the decoder never consults.

The failure is invisible to alignment metrics by construction. CAS_{fg} measures correlation between routing and ground truth, and correlation is compatible with an inert bottleneck. Worse, the alignment loss trains precisely what CAS_{fg} measures, so a high score confirms the router learned its supervision, not that the network uses it.

Why it happens. We rule out the obvious explanation: the experts have *not* collapsed to a shared function—they displace tokens by 59–74% of their norm and differ pairwise by 0.9. The cause is architectural, and it is two independent leaks that U-Net-style designs invite:

1. **Through the module.** The expert computes $h + c_k(h)$. The residual carries the full feature vector to the decoder whichever expert fires, so routing modulates the representation but never gates it.
2. **Around the module.** The decoder’s skip connections carry encoder features directly to the output. In our transformer backbone the CCR tap sits at stage 2 of 5, so the two *deeper*, more semantic stages bypass it entirely—the decoder can decide what tissue it is looking at without consulting the routing at all.

Either leak alone suffices to make routing inert, and neither is visible in CAS_{fg} .

Closing them. We gate both—the expert returns $c_k(h)$ alone, and a scalar α attenuates the bypassing skips—and sweep α to trace what a genuinely causal concept bottleneck costs. **[PENDING: one-sentence result of the sweep]**

Contributions.

1. **A diagnostic protocol for concept bottlenecks.** Three cheap tests—causal intervention, bottleneck ablation, and a linear-probe control—that distinguish a bottleneck the model uses from one it ignores. Alignment metrics cannot (Sec. 4).
2. **Evidence the naive design fails it.** A concept bottleneck with CAS_{fg} up to 0.96 and best-in-class concept discriminability is causally inert, and we localize the cause to two specific architectural leaks (Sec. 5.3).
3. **A repair, and its price.** Gating both leaks makes the routing **[PENDING: causally decisive / measurably more decisive]**, at a segmentation cost of **[PENDING: Dice delta]**. We report the full trade-off curve rather than a single operating point (Sec. 5.5).
4. **A probe control that reframes the baseline.** On an encoder trained with no concept supervision, a linear probe already separates the concepts at 0.81—above every post-hoc method. Concept-discriminability improvements over saliency baselines are therefore weaker evidence than the literature treats them as (Sec. 5.4).

2 Related Work

Post-hoc attribution. Grad-CAM [10], integrated gradients [12], SHAP [6] and attention rollout [1] all reconstruct a saliency map after prediction. On dense prediction the extension proceeds through heuristics, and the output is an input-*sensitivity* map rather than a concept assignment. We use four such methods as baselines and evaluate them under an identical protocol.

Intrinsic interpretability and concept bottlenecks. CBMs [5] and ProtoPNet [3] make the explanation part of the computation, for scalar-output classification. Neither addresses dense prediction, where the explanation must be per-voxel. Critically for this paper, the standard validation in this line is *concept accuracy or alignment*—does the concept layer predict the annotated concepts—which our results show is not sufficient. A line of work on *concept leakage* in CBMs shows that soft concept representations can smuggle task information past the bottleneck **[PENDING: cite the concept-leakage papers – Margeloiu et al. “Do Concept Bottleneck Models Learn as Intended?” and Mahinpei et al. “Promises and Pitfalls of Black-Box Concept Learning Models”; VERIFY venue/year before adding to references.bib]**; we report a complementary and more basic failure, in which information bypasses the bottleneck through the *architecture* rather than through the concept encoding. Positioning against that literature matters: leakage says the concept vector carries too much, whereas we find the concept vector is not read at all.

Mixture-of-experts routing. Sparse MoE [11,4,8] introduced routing for conditional computation. There, routing is causally decisive because the selected expert’s output *is* the layer output. Our results show that transplanting the mechanism into a skip-connected encoder–decoder silently voids that property, which is the assumption an interpretability claim would rest on.

3 The CCR Module

3.1 Router, experts, dispatch

Let $h_t \in \mathbb{R}^D$ be the bottleneck token at spatial position t . A router produces $p(t) = \text{softmax}(Wh_t/\tau + \beta \text{sim}(h_t, \Pi)) \in \Delta^{K-1}$, where Π are K learnable concept prototypes and τ is an annealed temperature. Each concept owns an expert E_k . Training uses soft dispatch, inference hard dispatch:

$$\text{soft: } \tilde{h}_t = \sum_k p_k(t) E_k(h_t), \quad \text{hard: } \tilde{h}_t = E_{k^*(t)}(h_t), \quad k^*(t) = \arg \max_k p_k(t). \quad (1)$$

$p(t)$ is the explanation: a per-region distribution over clinical concepts, read directly from the forward pass.

3.2 The two leaks

Eq. 1 is what an intrinsic explanation requires *only if* $\tilde{h}_t$ is the decoder’s sole source of concept information. Two standard design choices break that.

Residual (through). The natural expert is a pre-norm residual MLP, $E_k(h) = h + c_k(h)$, whose residual stabilizes early training. It also means the decoder receives h intact regardless of k^* . Routing then reweights a signal that arrives anyway. We expose this as a switch: $E_k(h) = c_k(h)$ makes the expert transform the only path. The standard near-zero initialization of c_k ’s output layer must be relaxed in this mode, since c_k is now the signal rather than a correction.

Skips (around). Encoder–decoder segmentation networks connect encoder stages directly to matching decoder stages. Every such connection is a path from input to output that does not pass the bottleneck. We attenuate them by a scalar α , sweeping from $\alpha=1$ (standard) to $\alpha=0$ (bottleneck is the only route). Skips carry spatial detail as well as semantics, so α trades boundary quality against causal control—the trade-off Sec. 5.5 measures.

A related placement issue is easy to miss. In a hierarchical transformer encoder the “bottleneck” is ambiguous, and inserting CCR at an intermediate stage leaves *deeper*, more semantic stages flowing around it. We report this because our own first design made exactly this error.

3.3 Objective and curriculum

We train with $\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{div}} + \lambda_3 \mathcal{L}_{\text{ent}} + \lambda_4 \mathcal{L}_{\text{bnd}}$, where $\mathcal{L}_{\text{align}}$ is a foreground-only focal BCE pushing $p_k(t)$ toward the ground-truth concept indicator, $\mathcal{L}_{\text{div}}$ decorrelates routing vectors, $\mathcal{L}_{\text{ent}}$ is a small entropy regularizer ($\lambda_3 \leq 0.01$), and $\mathcal{L}_{\text{bnd}}$ is a boundary term. A three-phase curriculum (warm-up / alignment / refinement) anneals τ from 2.0 to 0.5. Note that $\mathcal{L}_{\text{div}}$ acts on routing *vectors*, not on expert *functions*: nothing in the objective forces $E_a \neq E_b$. We verify functional divergence empirically rather than assuming it (Sec. 5.3).

4 A Diagnostic Protocol for Concept Bottlenecks

Alignment is necessary but not sufficient. We propose three tests, all cheap, that together establish whether a concept bottleneck is load-bearing.

Definitions. $\text{CAS}_{\text{fg}}(k) = \text{Pearson}(p_k(t), M_k(t))$ over foreground tokens measures routing/ground-truth *alignment*. $\text{DD}(k) = 1 - \text{Pearson}(P_k^{\text{dec}}, p_k)$ measures how far the decoder’s posterior departs from the routing.

Test 1: causal intervention. Replace $k^*(t)$ with a chosen b for the tokens assigned to a , changing nothing else, and re-run the decoder. Report the change in the posterior of class b over exactly those voxels, the fraction of their labels that flip to b , the change in *off-target* classes (specificity), and the change *outside* the intervened region (locality). The baseline must run through the same hard-dispatch path so the identity intervention $a \rightarrow a$ is a bit-exact no-op; then any off-diagonal effect is attributable to re-routing alone. *This test is impossible for post-hoc attribution: there is nothing in a saliency map to intervene on.*

Test 2: bottleneck ablation. Replace the module output with the raw encoder features (*bypass*) or with zeros (*zero*), and measure the Dice change. If destroying the bottleneck barely hurts, the decision does not live there and no amount of concept alignment will make it interpretable.

Test 3: linear-probe control. Fit multinomial logistic regression on the features *entering* the module and score it exactly as the router is scored. This asks whether the concept structure is produced by the concept machinery or merely read off features that already contained it. Run it on an ablated model trained without concept supervision for the strong version of the control.

5 Experiments

5.1 Setup

BraTS 2024 glioma [7,2]: four co-registered MRI modalities, labels remapped to {0:Background, 1:NCR, 2:Edema, 3:ET}, patient-wise 75/12.5/12.5 split ($n_{\text{test}}=168$),

Table 1. The naive bottleneck by standard metrics (BraTS test, $n=168$). Concept-discrim. is concept k vs. other tumor voxels (0.5=chance). By these measures the module works.

	Dice WT/TC/ET	CAS _{fg} N/E/T	Concept-discrim.
CCR-Net (Swin)	0.915/0.845/0.837	0.69/0.94/0.93	0.872
CCR-SegResNet (CNN)	0.927/0.883/0.869	0.80/0.96/0.95	0.883
Occlusion (best post-hoc)	—	—	0.741
Gradient saliency	—	—	0.601
Integrated gradients	—	—	0.583
Grad-CAM	—	—	0.521

z -scored per modality, 128^3 inputs, 80 epochs of AdamW (10^{-4} , cosine), mixed precision. We instantiate CCR on a Swin transformer encoder with a UNETR decoder (**CCR-Net**) and on a SegResNet CNN (**CCR-SegResNet**); the module, losses and curriculum are identical in both.

5.2 The naive bottleneck aligns

Table 1 reports the standard picture. Segmentation is competitive (0.927/0.883/0.869 Dice for the CNN), routing tracks the concepts closely (CAS_{fg} 0.80/0.96/0.95), and on concept-discriminability—concept k versus the *other* tumor voxels—CCR reaches 0.88 against 0.74 for the strongest post-hoc method, with integrated gradients and Grad-CAM near chance. Removing $\mathcal{L}_{\text{align}}$ collapses CAS_{fg} to ≈ 0 while Dice barely moves, which we read at the time as evidence that alignment is *learned*. It is; but the same number is the first hint of the problem, since it says the segmentation does not need the alignment.

5.3 ...and is inert

Intervention. Re-routing moves nothing (Table 2). The target-class posterior shifts by at most 0.002 and at most 0.2% of voxels change label. The effect is not concept-specific—off-target classes move *more* than the target in several cells (0.0052 vs. 0.0009)—and not localized, since voxels outside the intervened region move as much as those inside. The identity diagonal is exactly 0, so this is a property of the model and not of the instrumentation.

The experts are not degenerate. The obvious explanation—all experts learned the same function—is false. Experts displace tokens by 0.59–0.74 of their norm and differ pairwise by 0.84–1.09 (relative L_2). The routing selects genuinely different transformations; they simply do not reach the output.

Bottleneck ablation. They do not reach the output because the output does not need them. Replacing the module with raw features or with *zeros* costs almost nothing: edema Dice $0.879 \rightarrow 0.865$, ET $0.862 \rightarrow 0.806$, NCR $0.256 \rightarrow 0.234$. A

Table 2. Causal intervention on CCR-Net. Re-routing the tokens assigned to the row concept through the column concept’s expert. *Left*: change in the target-class posterior over the re-routed voxels. *Right*: fraction of those voxels whose predicted label became the target. Diagonal is the identity control and is exactly zero by construction.

from ↓ / to →	Δ target posterior			label-flip fraction		
	NCR	Edema	ET	NCR	Edema	ET
NCR	0.0000	0.0003	0.0002	0.0000	0.0003	0.0003
Edema	0.0009	0.0000	0.0012	0.0008	0.0000	0.0007
ET	0.0018	0.0014	0.0000	0.0020	0.0014	0.0000

Table 3. The repair sweep (CCR-SegResNet). Causal control is the mean off-diagonal label-flip fraction from Test 1; higher means the routing decision determines the output.

residual	α	Dice WT/TC/ET	CAS _{fg} N/E/T	Causal control	Zero-ablation Δ Dice
V0 yes	1.0	0.927/0.883/0.869	0.80/0.96/0.95	0.002	0.014
V1 no	1.0	[PENDING: V1 Dice]	[PENDING: V1 CAS]	[PENDING: V1 control]	[PENDING: V1 ab
V2 no	0.5	[PENDING: V2 Dice]	[PENDING: V2 CAS]	[PENDING: V2 control]	[PENDING: V2 ab
V3 no	0.0	[PENDING: V3 Dice]	[PENDING: V3 CAS]	[PENDING: V3 control]	[PENDING: V3 ab

decoder that can segment without its bottleneck is a decoder that will ignore any concept structure placed there.

5.4 The probe control

A linear probe on the features entering the module scores 0.839 against the router’s 0.872, and *wins* on ET. Run on the model trained with no concept supervision at all, the probe still reaches 0.8115—above every post-hoc method in Table 1. The concept structure is largely a property of a competent segmentation encoder, not of the concept machinery. This has a consequence beyond our system: reporting that a concept module separates concepts better than saliency baselines is weak evidence, because a logistic regression on an unsupervised bottleneck clears that bar.

5.5 Closing the leaks

We now gate both leaks and sweep α . V0 is the naive configuration above; V1 removes the residual; V2 and V3 additionally set $\alpha=0.5$ and $\alpha=0$.

[PENDING: 2–4 sentences: where the curve bends; whether causal control rises materially before Dice degrades; which configuration is the recommended operating point and why]

Intervention after repair. [PENDING: the V3 (or best) intervention matrix, with the same specificity and locality columns as Table 2, showing whether re-routing now moves the prediction toward the target concept]

[PENDING: fig: causal control (x) vs Dice (y), four points V0–V3, annotated]

Fig. 1. The interpretability–accuracy exchange rate. Each point is a trained model. Moving right means the routing decision increasingly determines the prediction; moving down means segmentation degrades. V0 is the naive concept bottleneck: maximally accurate and causally inert.

5.6 Backbone transfer

The same module, unchanged, in a Swin transformer and a SegResNet CNN produces the same qualitative picture: strong alignment, near-zero causal control. [PENDING: whether the repair transfers to the transformer backbone as well]

6 Discussion and Limitations

What alignment metrics can and cannot show. CAS_{fg} and concept-discriminability measure correlation between an internal signal and annotated concepts. A module can maximize both while contributing nothing to the prediction, and our alignment loss trains precisely what these metrics score. Any claim that a concept layer makes a model interpretable requires an intervention.

No controlled accuracy comparison. We did not train plain-backbone counterparts under our split, so the accuracy cost of inserting the module in the V0 configuration is unquantified; the sweep in Sec. 5.5 measures *relative* cost across configurations, which is the quantity the trade-off argument needs.

Resolution. The bottleneck is 16^3 , so the explanation is coarse and necrotic core—tens of tokens per case—is the consistent weak spot across every metric.

Single dataset and $K=4$. Concept granularity follows the BraTS protocol; transition zones are forced into one concept. Whether the leak analysis generalizes to other anatomies and concept vocabularies is untested, though the two leaks are properties of the architecture family rather than of the data.

Scope of the explanation. CCR explains at the level of clinical concepts—which expert processed each region—not the router’s internal reasoning. [PENDING: if the repair succeeds: one sentence on what a clinician gains that the segmentation output does not already provide]

7 Conclusion

We set out to make the routing decision the explanation, and found that the obvious implementation produces a concept bottleneck that is beautifully aligned and causally inert: CAS_{fg} up to 0.96, best-in-class concept discriminability, and a prediction that does not move when the routing is overwritten. The cause is not degenerate experts but two architectural leaks—a residual path through the module and skip connections around it—that alignment metrics cannot see.

Closing them **[PENDING: result]**. Beyond this system, the finding is a caution for intrinsic interpretability in dense prediction: a concept layer is only an explanation if the model consults it, and demonstrating that requires intervening, not correlating.

22 unresolved \pending placeholders remain.

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